

Why combine HALPER and IPP?

Proposed strategy for 35-species liver ATAC data • Conceptual schematic, not measured results

A Common input

Exact genome assemblies + species alignment + source liver ATAC peak

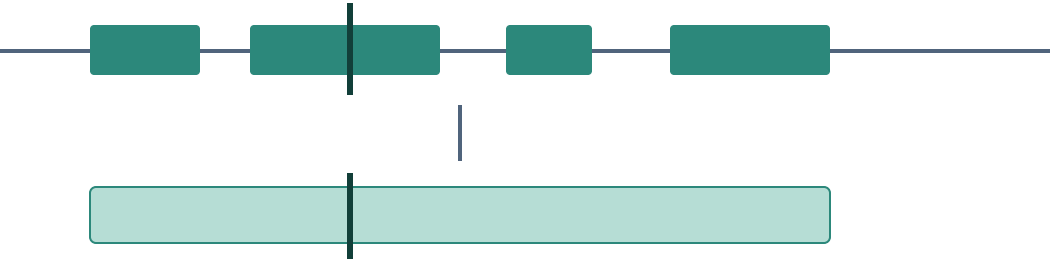
Use the same summit for both tools; keep stable source peak IDs.



Peak interval and same focal point

B1 HALPER: recover an interval

Mapped fragments + mapped summit

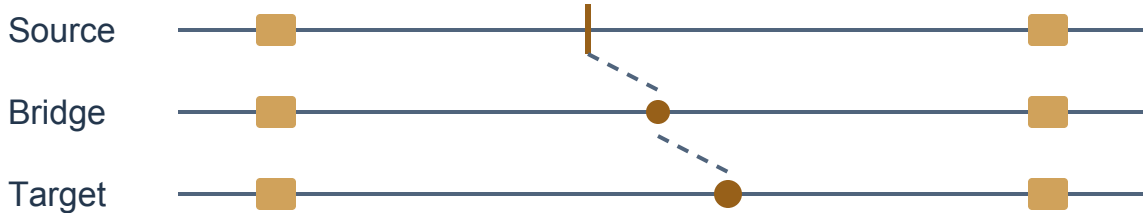


Strength: reconstructs fragmented correspondence.

Limit: focal mapping and interval filters must pass.

B2 IPP: estimate a corresponding point

Alignable anchors + optional bridging species



Strength: proposes loci beyond direct matches.

Limit: positional estimate; no enhancer boundaries.

C Combine evidence without collapsing uncertainty

Agreement

Same locus • retain both records

Tool-specific candidate

HALPER-only or IPP rescue

Conflict / multiple mappings

Retain alternatives • flag ambiguity

D Test biological conservation separately

Target liver ATAC replicates • local synteny • matched controls • histone marks • functional assays

Homologous locus ≠ conserved accessibility ≠ experimentally conserved regulatory function

Expected benefit: broader candidate recovery. Accuracy and extra recovery must be benchmarked.

Shared HAL input can produce correlated errors: agreement is not independent validation. See guide for sources and criteria.